Milestone 5 Report: Ensembling

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Milestone Report: Model Ensembling and Prediction Improvement

Objective

The objective of this milestone was to improve multiple-choice question prediction performance by combining the outputs of fine-tuned transformer models and exploring post-processing techniques. The work focused on extracting model logits, generating Top-3 predictions, applying ensemble methods, and evaluating prediction quality using MAP@3.

1. Extracting and Sorting Logits for Top-3 Predictions

Both the fine-tuned DeBERTa and RoBERTa models produce a vector of logits representing the confidence for each of the five answer choices (A, B, C, D, and E).

The following steps were performed:

- Obtained logits from each model after inference.
- Applied the Softmax function to convert logits into probability distributions.
- Sorted the probabilities in descending order using NumPy.
- Selected the three highest-probability answer options.
- Converted the indices back to answer labels (A–E).
- Formatted the predictions in the Kaggle submission format (e.g., A C B).

This approach ensured that each prediction included the three most likely answers rather than only the highest-confidence answer, which is required for MAP@3 evaluation.

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2. Ensemble Techniques for Combining Multiple Models

To improve prediction robustness, outputs from two independently fine-tuned transformer models were combined.

Simple Probability Averaging

The probability distributions from DeBERTa and RoBERTa were averaged using:

$$
P_{ensemble}=\frac{P_{DeBERTa}+P_{RoBERTa}}{2}
$$

The averaged probabilities were then sorted to generate the final Top-3 prediction.

Weighted Probability Averaging

Since DeBERTa generally performed better on the validation data, higher importance was assigned to it.

The weighted ensemble used:

$$
P_{final}=0.7\times P_{DeBERTa}+0.3\times P_{RoBERTa}
$$

This method preserved DeBERTa's stronger predictions while still benefiting from RoBERTa's complementary information.

The weighted ensemble was used to:

- Generate Top-1 predictions.
- Produce Top-3 Kaggle submission strings.
- Create the final submission.csv.
- Compare predictions against the standalone DeBERTa model.
- Measure confidence gain after ensembling.

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3. Test-Time Augmentation (TTA)

A simple Test-Time Augmentation strategy was implemented to improve prediction stability.

For each question, two versions of the input were created:

1. Original prompt.
2. Prompt prefixed with:

"Answer the following multiple-choice question carefully:"

The model performed inference on both versions.

The resulting probability distributions were averaged:

$$
P_{TTA}=\frac{P_{original}+P_{augmented}}{2}
$$

The averaged probabilities were then used to determine the final prediction.

The effect of TTA was analyzed by counting how many Top-1 predictions changed compared to using only the original prompt.

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4. Confidence Analysis

Prediction confidence was measured as the maximum probability assigned to any answer option.

For the first 100 samples, the following values were computed:

- DeBERTa confidence.
- Weighted ensemble confidence.

Confidence gain was calculated as:

$$
Confidence\ Gain=Confidence_{Ensemble}-Confidence_{DeBERTa}
$$

The number of samples with positive confidence gain was recorded to evaluate whether ensembling increased prediction certainty.

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5. Prediction Comparison

Several comparisons were carried out to understand the effect of ensembling:

- **Comparison of Top-1 predictions between DeBERTa and the weighted ensemble.**
- **Comparison of ordered Top-3 prediction strings.**
- **Measurement of how many predictions changed after ensembling.**
- **Evaluation of confidence improvements.**

These analyses helped determine whether combining multiple models produced more reliable predictions than using a single model.

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6. MAP@3 Evaluation

The weighted ensemble was evaluated on the validation dataset using the Mean Average Precision at 3 (MAP@3) metric.

The evaluation process involved:

- **Generating Top-3 predictions for each validation sample.**
- **Comparing the ranked predictions with the correct answer.**
- **Assigning partial credit based on the rank of the correct answer.**
- **Averaging the scores across all evaluated samples.**

MAP@3 provides a more informative assessment than Top-1 accuracy because it rewards models that rank the correct answer highly, even if it is not the first prediction.

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7. Strategies for Further Improving Predictions

Several additional techniques can be explored to improve prediction performance:

- **Fine-tuning larger transformer models such as DeBERTa-v3-base or DeBERTa-v3-large.**
- **Combining outputs from more than two models using advanced ensemble methods.**
- **Optimizing ensemble weights through validation experiments instead of fixed values.**
- **Applying additional prompt-based Test-Time Augmentation techniques.**
- **Using cross-validation to create stronger ensemble models.**

- **Performing probability calibration to obtain more reliable confidence estimates.**
- **Using knowledge retrieval or Retrieval-Augmented Generation (RAG) to provide external context for difficult questions.**
- **Increasing training data through data augmentation and improved preprocessing.**

Conclusion

This milestone successfully demonstrated how prediction quality can be improved using probability-based ensembling and Test-Time Augmentation. Logits were converted into probability distributions, Top-3 predictions were generated in Kaggle format, and weighted ensembles combining DeBERTa and RoBERTa produced more robust predictions. Additional analyses, including confidence comparison and MAP@3 evaluation, provided insight into the effectiveness of these techniques and established a strong foundation for further improvements in multiple-choice question answering systems.